KAIZEN MATHS

Lesson starter

Indices and Exponent Laws

Complex Multi-Step Index Simplification

Name _____

S1

Simplify: $a^3 \times a \div a^2 \times a$

S2

Simplify: $((p^3)^3 \times p^5) \div p^2$

S3

Simplify: $2^3 \times 2 \div 2^4 \times 2$

S4

Simplify: $((2^4)^3 \times 2^3) \div 2^4$

Show your working

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ALG-934A-L3 · S

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Complex Multi-Step Index Simplification

Name _____

E1

Simplify: $y^2 \times y \div y^2 \times y^3$

E2

Simplify: $((a^4)^2 \times a^3) \div a^4$

E3

Simplify: $x^4 \times x^2 \div x^4 \times x^2$

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Complex Multi-Step Index Simplification

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E1

Simplify: $y^2 \times y \div y^2 \times y^3$

E2

Simplify: $((a^4)^2 \times a^3) \div a^4$

E3

Simplify: $x^4 \times x^2 \div x^4 \times x^2$

ALG-934A-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Complex Multi-Step Index Simplification

Name _____

E1

Simplify: $y^2 \times y \div y^2 \times y^3$

E2

Simplify: $((a^4)^2 \times a^3) \div a^4$

E3

Simplify: $x^4 \times x^2 \div x^4 \times x^2$

ALG-934A-L3 · E

Show your working

KAIZEN MATHS

Exit ticket

Indices and Exponent Laws

Complex Multi-Step Index Simplification

Name _____

E1

Simplify: $y^2 \times y \div y^2 \times y^3$

E2

Simplify: $((a^4)^2 \times a^3) \div a^4$

E3

Simplify: $x^4 \times x^2 \div x^4 \times x^2$

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Show your working

Teacher answers & working

Indices and Exponent Laws · Complex Multi-Step Index Simplification

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S1

Simplify: $a^3 \times a \div a^2 \times a$

Answer:

a^3

Working:

$a^3 \times a = a^4$

$a^4 \div a^2 = a^2$

$a^2 \times a = a^3$

$a^3 \times a \div a^2 \times a = a^3$

S2

Simplify: $((p^3)^3 \times p^5) \div p^2$

Answer:

p^{12}

Working:

Work from the innermost brackets first.

$(p^3)^3 = p^9$

$p^9 \times p^5 = p^{14}$

$p^{14} \div p^2 = p^{12}$

$((p^3)^3 \times p^5) \div p^2 = p^{12}$

S3

Simplify: $2^3 \times 2 \div 2^4 \times 2$

Answer:

2

Working:

$2^3 \times 2 = 2^4$

$2^4 \div 2^4 = 1$

$1 \times 2 = 2$

$2^3 \times 2 \div 2^4 \times 2 = 2$

Teacher answers & working

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S4

Simplify: $((2^4)^3 \times 2^3) \div 2^4$

Answer:

2^{11}

Working:

Work from the innermost brackets first.

$(2^4)^3 = 2^{12}$

$2^{12} \times 2^3 = 2^{15}$

$2^{15} \div 2^4 = 2^{11}$

$((2^4)^3 \times 2^3) \div 2^4 = 2^{11}$

E1

Simplify: $y^2 \times y \div y^2 \times y^3$

Answer:

y^4

Working:

$y^2 \times y = y^3$

$y^3 \div y^2 = y$

$y \times y^3 = y^4$

$y^2 \times y \div y^2 \times y^3 = y^4$

E2

Simplify: $((a^4)^2 \times a^3) \div a^4$

Answer:

a^7

Working:

Work from the innermost brackets first.

$(a^4)^2 = a^8$

$a^8 \times a^3 = a^{11}$

$a^{11} \div a^4 = a^7$

$((a^4)^2 \times a^3) \div a^4 = a^7$

Teacher answers & working

Indices and Exponent Laws · Complex Multi-Step Index Simplification

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E3

Simplify: $x^4 \times x^2 \div x^4 \times x^2$

Answer:

x^4

Working:

$x^4 \times x^2 = x^6$

$x^6 \div x^4 = x^2$

$x^2 \times x^2 = x^4$

$x^4 \times x^2 \div x^4 \times x^2 = x^4$